

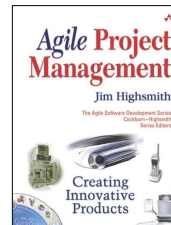
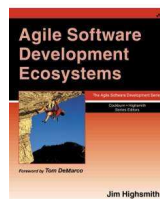


An Adaptive Performance Management System

Jim Highsmith

Director, Agile Project Management Practice

Fellow, Cutter Consortium



© Jim Highsmith

Building an Agile Enterprise

“Doubt is not a pleasant state, but certainty is a ridiculous one.” – Voltaire

Creating a competitive monster



Introductory Stories

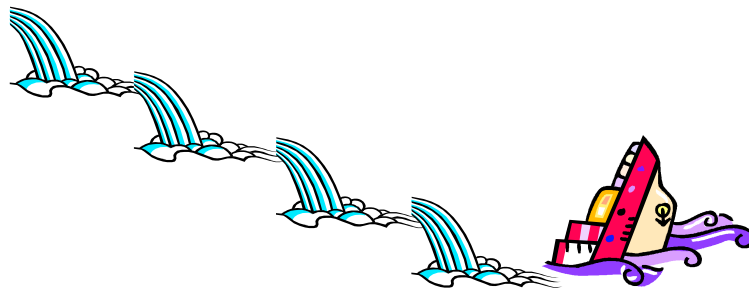
- Recent *BusinessWeek*-Boston Consulting Group survey showed that 72% of the senior executives in the survey named innovation as one of their top three priorities.
- “Experimentation matters because it is through learning equally what works and what doesn’t that people develop great new products, services, and entire businesses. But in spite of the lip service that is paid to “testing” and “learning from failure,” today’s organizations, processes, and management of innovation often impede experimentation.”

Stefan Thomke, *Experimentation Matters: Unlocking the Potential of New Technologies for Innovation*, Harvard Business School Press, 2003.



Introductory Stories

- In *Artful Making*, Harvard Business School Professor and Cutter Fellow Rob Austin and coauthor Lee Devin discuss a US \$125 million IT project disaster in which the company refused to improvise and change from the detailed plan set down prior to the project’s start. “*Plan the work and work the plan*” was their implicit mantra. And it led them directly to a costly and destructive course of action.... We’d all like to believe that this kind of problem is rare in business. It’s not.”



Introductory Stories

"I recently asked a colleague [CIO] whether he would prefer to deliver a project somewhat late and overbudget but rich with business benefits or one that is on time and underbudget but of scant value to the business. He thought it was a tough call, and then went for the on-time scenario. Delivering on time and within budget is part of his IT department's performance metrics. Chasing after the elusive business value, over which he thought he had little control anyway, is not."



Cutter Sr. Consultant Helen Puksza

Introductory Stories

- Standish Group "Chaos Reports"
 - 1994 — 82% challenged or failures, 2001—72% challenged or failures
- Definition of "success"
 - successful — the project is completed on time and on budget, with all the features and functions originally specified;
 - challenged — the project is completed and operational, but over budget, late, and with fewer features and functions than initially specified; and
 - failed — the project is canceled before completion or never implemented.
- "A complex system designed from scratch never works (Gall)."
- Where are OUTCOMES in this definition? When we play into the traditional measures of "success," we obscure the benefits of agile and damage our chances of creating a competitive monster.

Complete, Fixed Requirements are a Myth

Thomke & Reinertsen — “Agile Product Development,” California Management Review, 1998

- Out of hundreds of projects, no case in which requirements remained stable through design
- Of over 200 developers, fewer than 5% had complete requirements before design
- On average, design commenced with only 58% of requirements specified

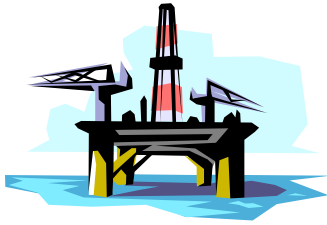
If you build the initially “specified” system, your project will be a FAILURE, because you didn’t LEARN anything!

Conclusions

- Innovation is critical to business success and agility is key to innovation.
- Management processes & performance measures restrict agility and innovation.
- Conformance to Plan is the norm for measuring success.
- Schedule is more important than customer value (outcome).

Problem Domains: Exploration versus Production

Exploration Drilling (Adaptation)



Production Drilling (Optimization)



Project Exploration (Innovation) Factor (EF)

Product Requirements Dimension	Product Technology(s) Dimension			
	Bleeding Edge	Leading Edge	Familiar	Well known
Erratic	10	8	7	7
Fluctuating	8	7	6	5
Routine	7	6	4	3
Stable	7	5	3	1

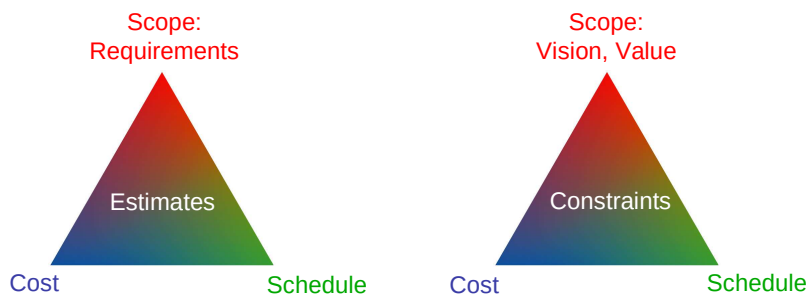
“If a project has no risks, don’t do it.”

Tom DeMarco & Tim Lister,

Waltzing with Bears: Managing Risk on Software Projects

Project & Product Management

- Identify Exploration Projects
- Manage them Differently
- Measure their Success Differently



So,

- To build agile organizations we **MUST** change how we measure performance.
- We must acknowledge that our performance measurement system impacts agility and our ability to innovate.
- We must alter our obsession with time to an obsession for outcomes, that is, customer value.
- We must separate the *outcome* performance measurements from the *output* performance measurements.

Conceptual Background Sources

- *Beyond Budgeting: How Managers Can Break Free from the Annual Performance Trap* (2003), Jeremy Hope and Robin Fraser.
- *Measuring and Managing Performance in Organizations* (1996), Rob Austin.

Conceptual Background

- *Beyond Budgeting: How Managers Can Break Free from the Annual Performance Trap* (2003), Jeremy Hope and Robin Fraser:
- “Budgets have since been hijacked by a generation of financial engineers that have used them as remote control devices to ‘manage by the numbers.’ They have turned budgets into *fixed performance contracts* that force managers at all levels to commit to delivering specified financial outcomes, even though many of the variables underpinning those outcomes are beyond their control.”

Budgeting Problems

- They are cumbersome and expensive.
 - “Some project leaders figure they have saved 95 percent of the time that used to be spent on budgeting and forecasting.”
 - “A vast area of well documented ignorance,” Ken Orr.
- They are out of kilter with the competitive environment.
 - “It is when the gap between key budget assumptions and emerging reality widens to the point at which the two bear little relation to each other that the problems begin.” Hope & Fraser
- Gaming the numbers is rampant.
 - Hiring criteria for budget analyst (project manager).

Beyond Budgeting—Principles

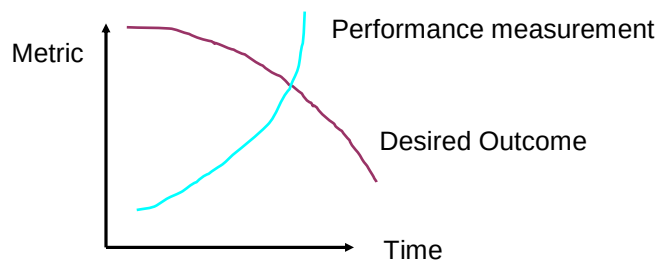
1. Provide a governance framework based on clear principles and boundaries.
2. Create a high-performance climate based on relative success.
3. Give people freedom to make local decisions that are consistent with governance principles and the organization's goals.
4. Place the responsibility for value-creating decisions on frontline teams.
5. Make people accountable for customer outcomes.
6. Support open and ethical information systems.

Conceptual Background

- *Measuring and Managing Performance in Organizations* (1996), Rob Austin:
- *"If there is a single message that comes from this book, it is that trust, honesty, and good intentions are more efficient in many social contexts than verification, guile, and self-interest."*

Austin—How Measurement Systems become Dysfunctional

- Step 1: Measurement system installed.
- Step 2: Performance tends to improve while people figure out the system.
- Step 3: People, under pressure, focus on measurement goals rather than outcomes (there is always a disconnect between the desired outcome and the measurement. Examples: (1) productivity; lines of code per day, (2) Compensation based on CMMi level).
- Step 4:



Austin—Dysfunctional Measurement Systems

- Furthermore, *“the implementation of more sophisticated measures caused more sophisticated dysfunctional reactions.”* Those being measured “can increase their rewards if they can successfully obliterate the correlation between true output and measured performance.”
- *“the ideal to which organizations should aspire is one in which workers are internally motivated and measurement provides them with self-assessment information. Measurement and motivation are decoupled. The challenge for managers is to become more trusting, able to inspire and communicate, and willing to help rather than be helped.”*

APMS—Goals

- To focus any enterprise group (project team, department, division, or company) on a set of desired strategic or tactical outcomes.
- To encourage those groups (project teams, etc.) to perform at a high level — measures of output.

APMS—Guidelines

- Build the measurement system on a foundation of trust, honesty, and an intent to increase organizational value.
- Place the most emphasis on measuring outcomes, not outputs, even if the metrics are not as easy to obtain, nor as precise.
- Create outcome metrics whose constraint metrics are as broad as possible (tolerance for variations) in order to encourage agility.
- Create output informational metrics that support people's innate internal motivational needs and provide them with aggregate measures of overall, rather than individual, performance.

Outcome Performance Metrics

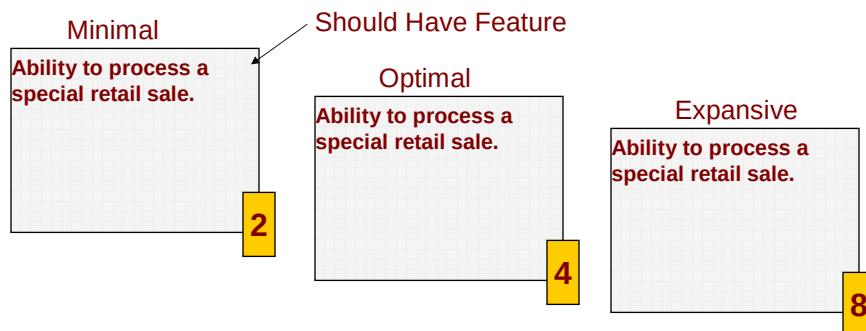
- “We increase return on investment by making continuous flow of value our focus,” Declaration of Interdependence (DOI).
- Outcomes are a project community responsibility.
“We deliver reliable results by engaging customers in frequent interactions and shared ownership (DOI).”
- Outcomes are business related.
 - IRACIS—Increase Revenue, Avoid Cost, Improve Service
 - Value of features/stories delivered
 - Value may be linked to reduced risk
 - Basis for decision making, not just reporting

Outcome Performance Metrics

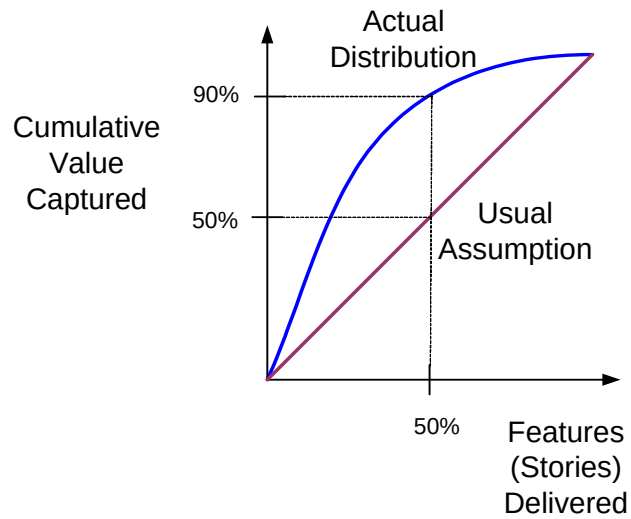
- We must alter our obsession with time to an obsession for outcomes, that is, customer value.
 - Fixing time as a constraint (time boxing) isn't the same as obsessing over schedule (fixing time enables us to focus on delivering value).
 - Time is a component of value.

Outcome Performance Metrics

- Capabilities, Features, Use Cases, Stories enable outcomes
 - They are NOT outcomes! Scope is NOT an outcome!
 - Variable Priority—MoSCoW (from DSDM); Must have; Should have; Could have; Won't have
 - Variable Implementation—Minimal, Optimal, Expansive (requirements & design)

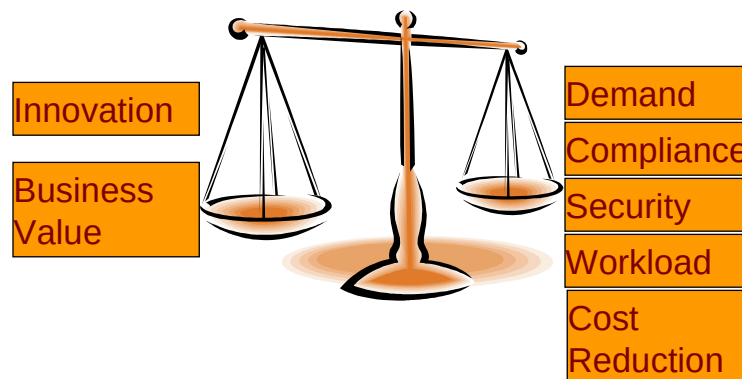


It's Not About Features, It's About Value

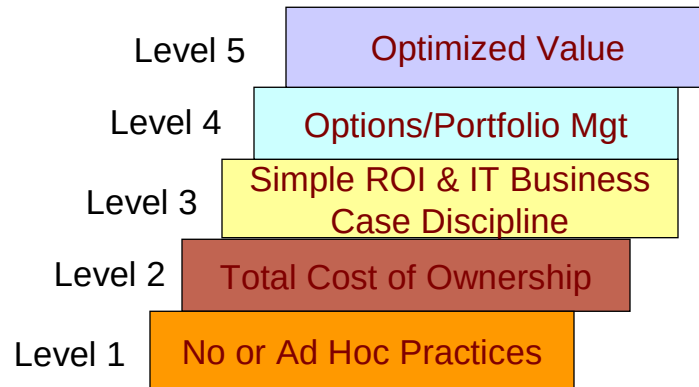


The New Challenge

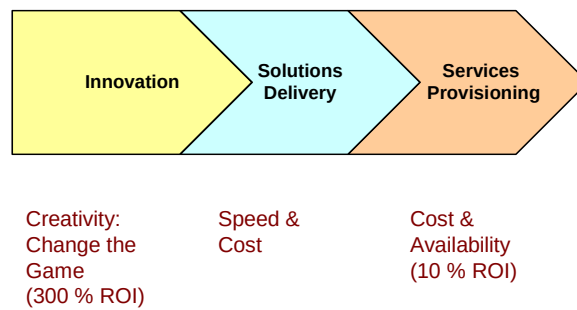
Martin Curley, Director IT Innovation, Intel Corporation



Business Value Maturity Model (Intel)



The IT Value Chain (Intel)



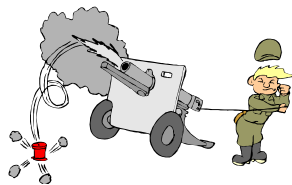
Intel's 17 Standard Measures of Value

- Days of inventory reduction
- Days of receivables outstanding
- Headcount reduction
- Headcount productivity
- Headcount productivity
- System end-of-life
- Materials discounts
- Capital, hardware, and software avoidance
- Unit cost avoidance
- Factory uptime
- Scrap reduction
- Risk avoidance
- Time-to-Market
- Opening new markets
- Optimizing existing markets
- Cross selling
- Vendor-of-choice

Intel White Paper: *Defining the Value of E-business: Seventeen Standard Measures*

Outcome Performance Metrics—Constraints

- Scope, Schedule, Cost, Quality are not Outcomes, but CONSTRAINTS
 - “A constraint is a measured characteristic of a project that if projected to exceed its variance target would subject the project to immediate cancellation or major revision requiring sponsor and possibly other management approval.”
 - Agile projects need constraints, not predictions
- Agility enables us to to alter course in mid-project in order to deliver the desired outcomes, therefore:
- Tighter constraints, less room for agility, reduces the probability of desired outcomes.



A Caution about Variation

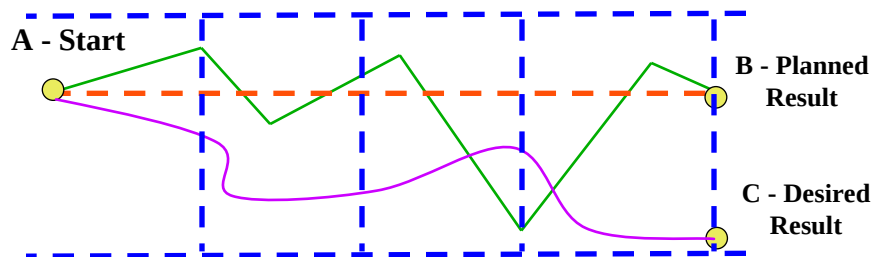
- Organizing for routine work: Drive out variation
- Organizing for innovative work: Encourage variation

David Anderson's blog

(<http://www.agilemanagement.net/Articles/Weblog/blog.html>)

Agile Planning & Execution

Conformance to Plan

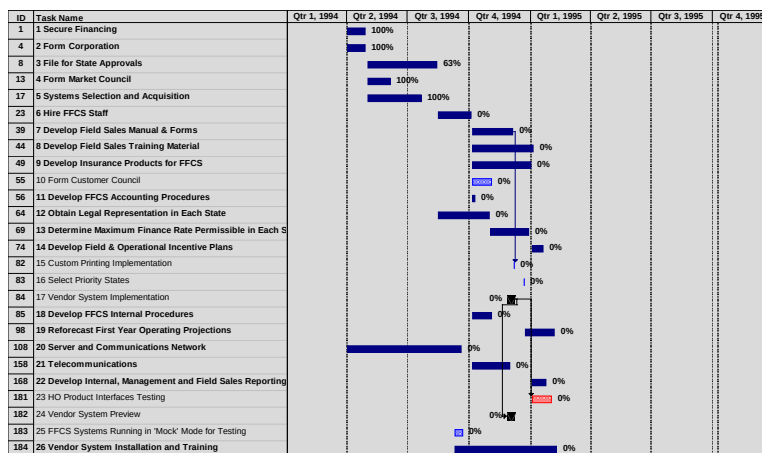


Conformance to Customer Outcomes Within Constraints

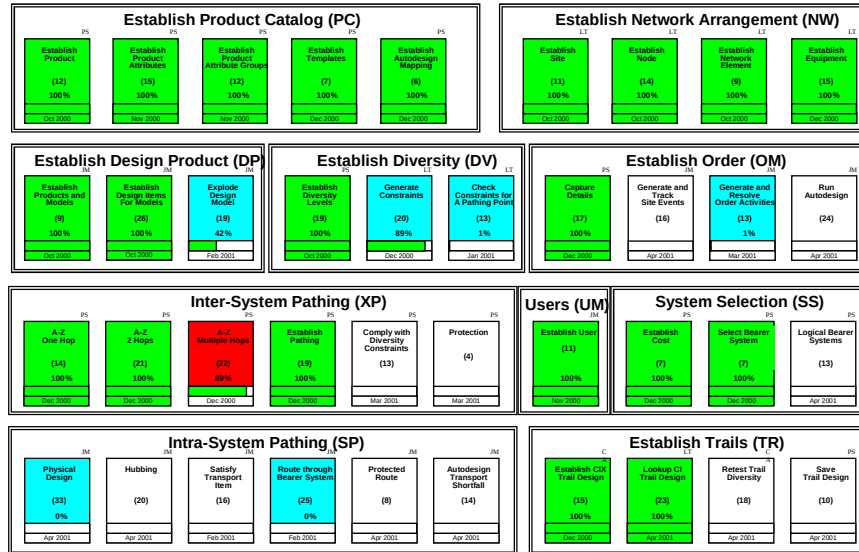
Trade-off Matrix for Constraints

	Fixed	Flexible	Acceptable
Scope			X
Schedule	X		
Resource			X
Quality		X	

Traditional Gantt Chart—What does it emphasize?



A Parking Lot Diagram—What does it emphasize?



©2007 Jim Highsmith

KEY:

Work In Progress

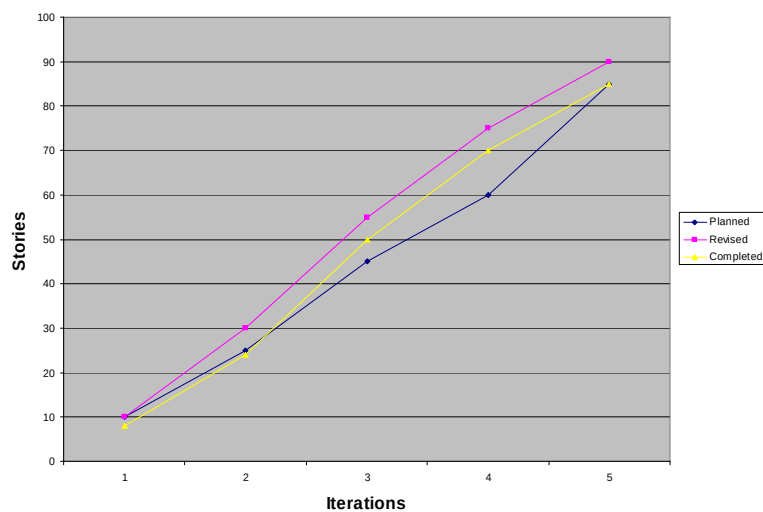
Attention

Full Completion

Progress Bar

Courtesy Jeff DeLuca

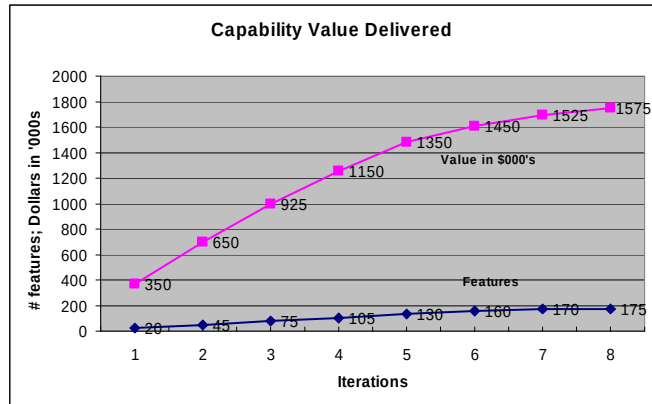
Story Performance



©2007 Jim Highsmith

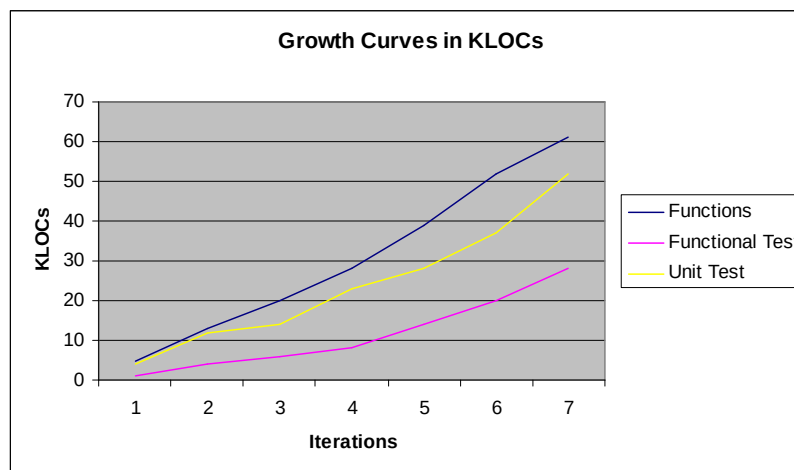
36

Capability Value Delivered

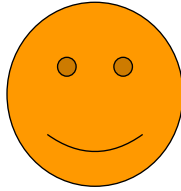


Denne, Mark, and Jane Cleland-Huang. *Software by Numbers: Low-Risk, High-Return Development*. Prentice Hall, 2003.

Metrics for Constraints: Quality



Outcomes and Constraints



Outcomes first:
business value.

Constraints second:
Scope, schedule, cost, quality



Agile Reporting

Transparency can be a big problem.



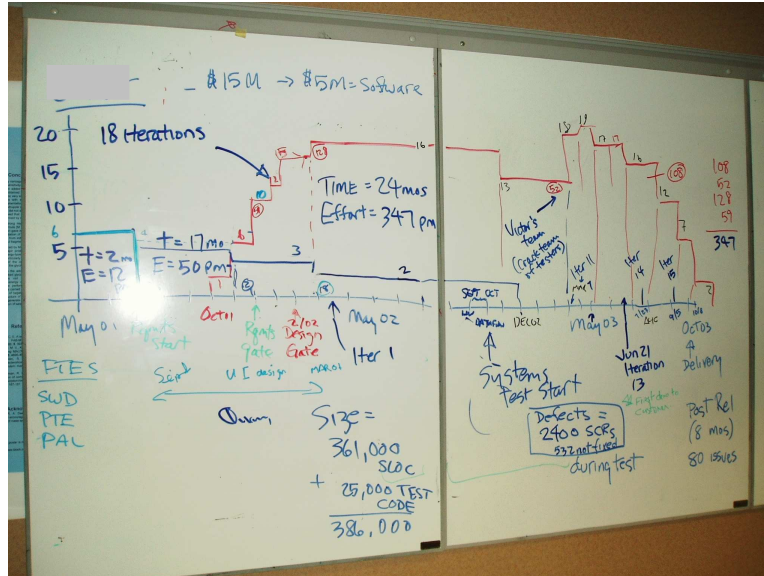
Output Performance Metrics

- Intentions are the key to success
- Give a team the data they need to get better
 - Compare its progress over time against itself — are we getting better?
 - Compare performance against other internal teams — how do we compare with others in our organization?
 - Compare performance against an external measure of like industries or projects — how do we compare with other companies?
- “Did we do the best we could do”, rather than “Did we meet the plan?”

Output Performance Metrics

- Five Core Metrics (Larry Putnam, SLIM Model)
 1. Quantity of function — i.e., scope, measured in terms of user stories, use cases, requirements, or features (depending on a particular situation. These may be measured ultimately as objects, modules, classes, or lines of code.)
 2. Productivity — expressed as functionality produced for the time and effort
 3. Time — the duration of the project in calendar months
 4. Effort — the amount of effort expended in person-months
 5. Reliability — as expressed in terms of defect rate
- Next 4 slides courtesy of Cutter Sr. Consultant Michael Mah.

Gathering Project Data

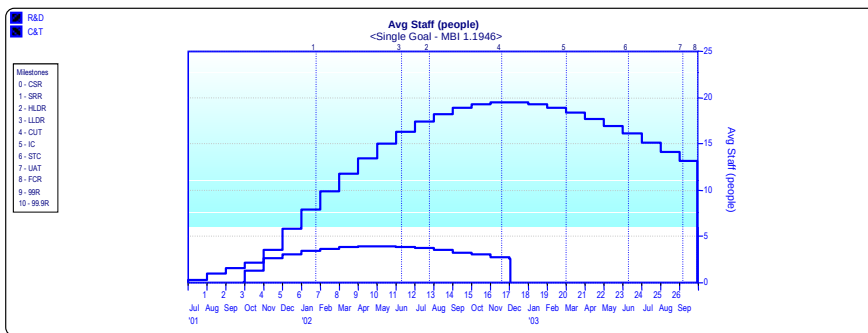


©2007 Jim Highsmith

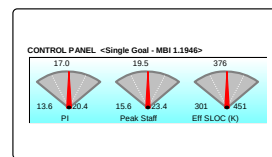
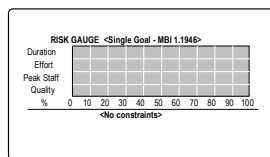
43

Project Model

Staffing and Probability Analysis



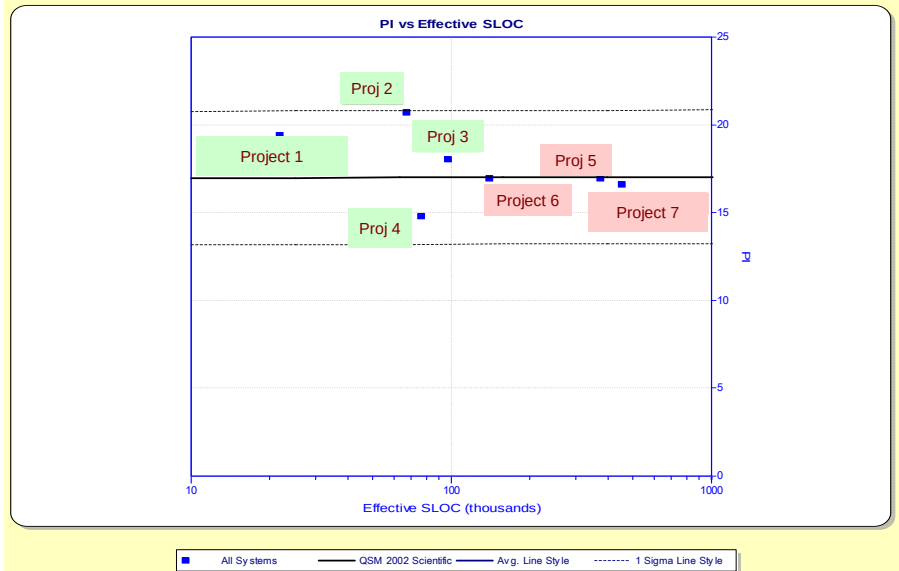
SOLUTION PANEL - Single Goal - MBI 1.1946			
	CAT	Life Cycle	
Duration	24.0	27.0	Months
Effort	367	397	PM
Cost	5887	6747	\$ (K)
Peak Staff	19.5	19.5	people
MTTD	0.3	0.3	Days
Start Date	9/29/2001	7/10/2001	
	PMI7.8	MBI12	EFF SLOC=376022



©2007 Jim Highsmith

44

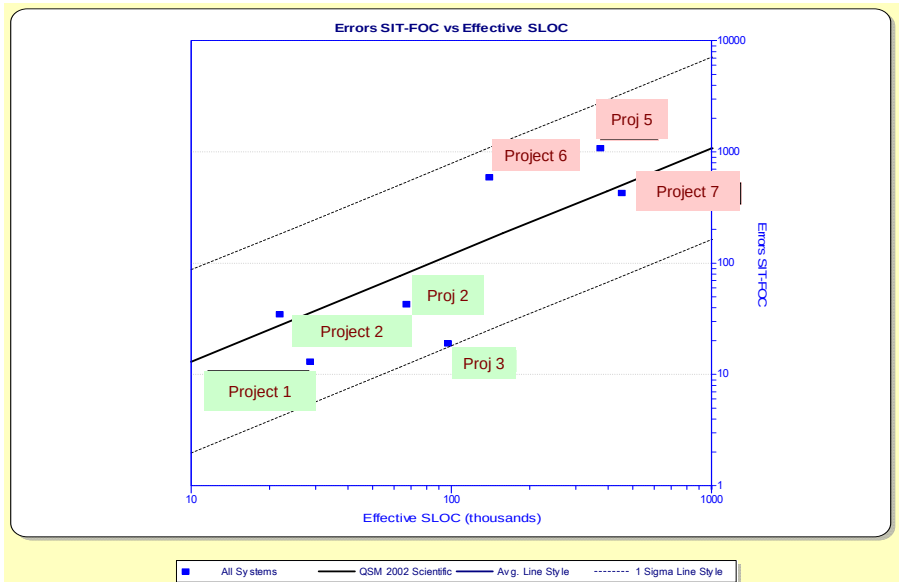
Trend Analysis — Productivity Index (PI)



©2007 Jim Highsmith

45

Trend Analysis — Defects During Test



©2007 Jim Highsmith

46

Improve Performance — Software for Scientific Instrument

- Overhaul the entire product development process
- Achieve software excellence (150,000 LOC project, a 2.5 point increase in the Productivity Index and defect reduction rates)

	Previous Performance	Current Performance	Percent Improvement
Project Cost	\$2.8 Million	\$1.1 Million	61%
Project Schedule	18 months	13.5 months	24%
Cumulative Defects	2,270	381	83%
Staffing	18	11	39%

Current & Project to Date Team Performance

	Productivity Index	Schedule	Reliability	Effort	Total	Average	% of Company
Team 4	5	3	4	4	16	4.0	110 to > 124%
Team 2	4	1	3	5	13	3.3	99 to > 109%
Team 1	2	3	4	1	10	2.5	75 to > 99%
Team 5	2	3	2	2	9	2.3	75 to > 99%
Team 3	1	2	3	1	7	1.8	< 75%

	Productivity Index	Schedule	Reliability	Effort	Total	Average	% of Company	Last Project to Date Performance	Performance Trend
Team 4	4	3	3	3	13	3.3	110 to > 124%	3.1	0.2
Team 2	3	2	3	4	12	3.0	99 to > 109%	2.7	0.3
Team 1	2	2	3	2	9	2.3	75 to > 99%	2.6	-0.4
Team 5	3	3	2	2	10	2.5	75 to > 99%	2.4	0.1
Team 3	3	2	3	2	10	2.5	< 75%	2.7	-0.2

External Industry Comparisons (Project-to-Date)

	Productivity Index	Schedule	Reliability	Effort	Average	% of Industry
Team 4	5	4	4	5	4.5	110 to > 124%
Team 2	5	3	4	5	4.3	110 to > 124%
Team 1	4	4	5	2	3.8	99 to > 109%
Team 5	4	4	3	3	3.5	99 to > 109%
Team 3	3	3	4	2	3.0	99 to > 109%

Remember

- Both Outcomes and Outputs must be considered
 - A very difficult project with good outcomes may have lower outputs
- Outcome measures are a shared responsibility
- Output measures are used by teams to improve themselves
- All measurement systems become dysfunctional over time

An Adaptive Performance Measurement System

- To build agile organizations we **MUST** change how we measure performance.
- We must acknowledge that our performance measurement system impacts agility.
- We must alter our obsession with time to an obsession for outcomes, that is, customer value.
- We must separate the *outcome* performance measurements from the *output* performance measurements.

Resources

- *An Adaptive Performance Management System*, Jim Highsmith, a Cutter Agile Project Management Executive Report. Download your free copy at www.cutter.com/offers/adaptiveperformance.html.
- *Beyond Budgeting: How Managers Can Break Free from the Annual Performance Trap* (2003), Jeremy Hope and Robin Fraser.
- *Measuring and Managing Performance in Organizations* (1996), Rob Austin.
- *Value-Based Software Engineering: Overview and Agenda*. Barry Boehm. University of Southern California CSE paper, Feb. 2005.